

READING PASSAGE 1

You should spend about 20 minutes on **Questions 1–13**, which are based on Reading Passage 1 below.

Tunnelling under the Thames

The first tunnel ever to be built under a major river was the tunnel under London's River Thames

At the beginning of the 19th century, the port of London was the busiest in the world. Cargoes that had travelled thousands of miles and survived all the hazards of the sea were unloaded on the banks of the Thames, only for their owners to discover that the most frustrating portion of their journey lay ahead. Consignments intended for the southern parts of Britain had to be lifted onto horse carts, pulled through the docks and across London Bridge, built in the 12th century and as impractical as its early date implies. By 1820, London Bridge had become the centre of the world's largest traffic jam.

It was an intolerable situation, and it was clear that if private enterprise could build another crossing closer to the docks, there would be good money to be made in tolls paid by users. Another bridge was out of the question, as this would deny sailing ships access to the city centre and ambitious men turned their thoughts to tunnelling beneath the Thames instead. This was not such an obvious idea as it might appear. Although increasing demand for coal had meant a great many tunnels had been dug in mines in Britain, working methods remained primitive. Tunnels were dug by men with simple tools, by candlelight. However, in 1807, a group of businessmen set themselves up as the Thames Archway Company. Their ambition was to tunnel below the Thames, but there was little to guide them as there had been no previous attempt to do this. Their chief engineer was Richard Trevithick, designer of the world's first high-pressure steam engine. His men made progress at the beginning, but then things began to go disastrously wrong, with muddy soil pouring into the tunnel. Eventually, the Thames Archway Company had had enough. Its funds were exhausted, Trevithick was sick from exposure to the river water, and its efforts had proved only that a passage under the river exceeded the limits of contemporary mining technology.

At that time, the only machines used in mines were pumps. It took a man of genius to recognise that a different sort of machine was needed, a machine that could prevent the roof and walls of a tunnel from collapsing. This man was Marc Brunel, a Frenchman who had become one of the most prominent engineers in Britain. Not long after the failure of the Thames Archway Company, Brunel saw a rotten piece of wood lying on the riverbank. Examining the wood through a magnifying glass, he observed it was infested with something that looked like a worm. Brunel realised that as it tunnelled through the wood, it would push chewed wood into its mouth and digest it, then excrete a hard substance that lined the new tunnel. Brunel realised that the worm's digging technique could be adapted to produce a new way of tunnelling.

His realisation led him to invent a device that has been used in one form or another in most major tunnels built since – the tunnelling shield. It consisted of a heavy iron frame that could be pushed forward a few inches at a time. The front of the frame was made up of a series of iron frames that could be folded back to allow miners to dig the ground ahead. Behind these frames was a wall consisting of a series of iron plates pressed against the tunnel face and supported on a set of horizontal wooden planks that would prevent the face from collapsing. It was a complex and rather cumbersome machine and not easy to use, but it seemed that it would protect the miners from the worst of the river's water. Brunel's team carefully examined earth samples taken from beneath the riverbed, and subsequently decided to dig the tunnel close to the muddy river bottom, where they could expect to find clay. This would be a more solid and safe substance to dig through than the sand that was found deeper down.

Brunel began work on his tunnel in 1825, but the problems of such an operation soon became apparent. Although the shield itself worked well, water began to drip into the tunnel. This was more of an annoyance than a danger while the pump was working, but this machine proved unreliable and sometimes failed altogether. When the pump broke down, work had to stop as the tunnel quickly flooded. There were occasions when the miners had to abandon their tools and flee for their lives. Even when Brunel's men were able to work, they had to run the constant risk of the pumps failing. They also complained of frequent headaches and dizziness, caused by the poor air quality. The air underground was dirty and stale, contaminated due to the lack of an adequate ventilation system. There were lighting problems too. Illuminating the tunnels by candlelight was a constant challenge. Lamps gave off only a very weak glow, and there were a number of accidents because the miners could not see what they were doing. Lastly, a number of Brunel's miners walked off the job because they could not tolerate the excessive temperatures that developed in the cramped conditions underground.

Despite all these setbacks, the tunnel finally emerged on the opposite river bank on August 12, 1841. Brunel's triumph, however, was only partial. The small payment per person made by the thousands of visitors who flocked to see the marvel hardly paid even a penny per foot of the tunnel's construction costs. Brunel had gone bankrupt long before the project was completed, and the government loan he had required to complete the project had to be paid back with interest. As a result, there was not enough funding to make it accessible to horse-drawn vehicles, as intended. Instead, the passageways were filled with souvenir sellers and entertainers. In the end, the tunnel was closed two years later, used at night, before it was finally closed entirely and fell into dereliction for decades.

It was only when the underground railway came to London in the 1880s that the Thames Tunnel found and achieved a measure of real usefulness. It was bought in 1869 by the East London Railway, who found it to be in such excellent condition that it was immediately pressed into service as a route for passenger trains heading east. The tunnel became, and remains, part of the London Underground network.

Questions 1–7

Do the following statements agree with the information given in Reading Passage 1?

In boxes 1–7 on your answer sheet, write

TRUE	<i>if the statement agrees with the information</i>
FALSE	<i>if the statement contradicts the information</i>
NOT GIVEN	<i>if there is no information on this</i>

- 1 In the early 19th century, the port of London was considered a safer destination than other ports.
- 2 London Bridge provided quick access for cargo being sent to southern Britain.
- 3 It was generally believed that a new river crossing would be profitable.
- 4 Building a second bridge crossing was initially considered to be the best solution.
- 5 It was believed that coal could be found under the River Thames.
- 6 The Thames Archway Company was the first group to try tunnelling below the Thames.
- 7 Some of Trevithick's men were injured during a mudslide at his tunnel.
- 8 The Thames Archway Company ran out of money to finance the tunnel project.

Questions 9–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 9–13 on your answer sheet.

Marc Brunel's tunnel

Preparing to build the tunnel

- Brunel noticed how a kind of **9** _____ made its tunnels in wood.
- Brunel created a device called a tunnelling shield, to protect people working under the river.
- Brunel planned to build a shallow tunnel so the earth would have a higher content of **10** _____.

Problems faced by miners

- There were frequent floods caused by mechanical breakdowns.
- The miners suffered from **11** _____ because of pollution in the tunnels.
- Lighting problems led to several **12** _____.
- Some workers quit because of the high temperatures in the tunnel.

After the tunnel was finished

- The tunnel was finally completed in 1841.
- Brunel did not have enough money to repay his debt to the **13** _____.
- The tunnel was abandoned until the 1880s.

判断题 (Questions 1–8)

题号	答案	题干翻译	精准定位 (英文原句)	定位译文	详细解释
1	NOT GIVEN	19世纪初, 伦敦港被认为比其他港口更安全。	P1 只说 “the port of London was the busiest in the world.”	仅说明“伦敦港是世界上最繁忙的港口”。	文中谈“最繁忙”“拥堵”, 未谈“更安全/safer”。无信息支持或反驳“更安全”, 故 NG 。
2	FALSE	伦敦桥为运往英国南部的货物提供了快捷通道。	P1 “London Bridge... as impractical as its early date implies... By 1820, London Bridge had become the centre of the world's largest traffic jam.”	伦敦桥“不切实际”, 1820 年成了“世界上最大的交通堵塞中心”。	题干称“快捷”, 原文强调“极度拥堵且不实用”, 与之相反, 故 FALSE 。
3	TRUE	人们普遍认为新的过河通道会盈利。	P2 “it was clear that if private enterprise could build another crossing... there would be good money to be made in tolls paid by users.”	“很明显, 如果私人企业能建另一处过河通道, 就能通过收取通行费赚到不少钱。”	原文直述“很明显...会赚钱 (good money)”, 等价于“普遍认为/公认会盈利”, 故 TRUE 。
4	FALSE	建第二座桥最初被认为是最佳方案。	P2 “Another bridge was out of the question, as this would deny sailing ships access to the city centre...”	“再建一座桥是不可能的, 因为会挡住帆船进城。”	原文明确说“out of the question (不在考虑之列)”, 与“最佳方案”相反, 故 FALSE 。
5	NOT GIVEN	人们认为泰晤士河底下能找到煤。	P2 “increasing demand for coal had meant a great many tunnels had been dug in mines in Britain” (仅此与煤有关)	只是说“英国因对煤的需求增加, 而在矿山挖了很多隧道”。	这里提煤只是社会背景, 未说“河下有煤”或“有人这样认为”。缺失该信息, 故 NG 。
6	TRUE	泰晤士拱道公司是第一批尝试在泰晤士河下挖掘的团体。	P2 “Their ambition was to tunnel below the Thames, but there was little to guide them as there had been no previous attempt to do this.”	他们想在泰晤士河下挖, 但“此前从未有人尝试过”。	“此前没有尝试”=他们是第一批尝试者, 故 TRUE 。
7	NOT GIVEN	特里维西克的一些工人在他隧道的泥石流中受伤。	P2 “things began to go disastrously wrong, with muddy soil pouring into the tunnel... Trevithick was sick from exposure to the river water”	出现“泥土涌入”, 并说“特里维西克病了”。	仅提到泥土涌入与 Trevithick 生病, 未提工人受伤。信息缺失, 故 NG 。
8	TRUE	泰晤士拱道公司因资金耗尽而无法继续资助工程。	P2 “Its funds were exhausted...”	“公司的资金被耗尽。”	题干即“ran out of money”, 与“funds were exhausted”完全同义, 故 TRUE 。

笔记填空 (ONE WORD ONLY, Questions 9–13)

题号	答案	题干翻译要点	精准定位 (英文原句)	定位译文	解释
9	worm	布鲁内尔注意到一种____如何在木头里打洞。	P3 “he observed it was infested with something that looked like a worm... as it tunnelled through the wood...”	他观察到木头里有“看起来像虫的东西”, 它在木头里打“隧道”。	notes 里的“a kind of ____”对应“looked like a worm”。题型限制一词, 用 worm 。
10	clay	计划挖较浅的隧道, 使土壤含有更高的____含量。	P3 “decided to dig the tunnel close to the muddy river bottom... expect to find clay. This would be... safer... than the sand found deeper down.”	决定靠近河底开挖, 以找到黏土, 比更深处的沙更稳更安全。	“较浅 (close to the bottom) → clay 多; 更深处是 sand”。填 clay 。
11	headaches	矿工因隧道污染而患上____。	P4 “They also complained of frequent headaches and dizziness, caused by the poor air quality.”	由于空气质量差, 他们经常抱怨头痛和眩晕。	“suffered from ____”与“complained of frequent headaches”搭配自然; 题型一词, 取 headaches (若考场给出 dizziness 亦可被接受, 本文按更常见搭配作答)。
12	accidents	照明问题导致多起____。	P4 “Lamps gave off only a very weak glow, and there were a number of accidents because the miners could not see what they were doing.”	灯光微弱, 因看不清发生了多起事故。	“a number of accidents”直接对应“多起事故”, 填 accidents 。
13	government	布鲁内尔无力偿还他对____所欠的债务。	P5 “the government loan he had required... had to be paid back with interest.”	他所借的政府贷款需要连本带息偿还。	“debt to the government (对政府的债务)”= 原文“government loan”。填 government 。